Chemical Engineering Part II

Equilibrium Thermodynamics

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[NOTE ON COLOUR CODING]

C19th – steam engines: heat/work, pure substance VLE

C20th – petrochemicals: property calcs in complex multicomponent mixtures (distillation)

C21st – diverse chemical products: complex multicomponent multiphase systems, phase diagrams

Introduction

Part IA and IB courses on Process Calculations & thermo:

- How to calculate properties for pure gases and liquids
- Equations of State (vdW, P-R)
- Criterion for equilibrium in a closed system - G
- Pure substance vapour-liquid equilibrium
- What is chemical potential, and how does it relate to G
- “Easy” mixtures using EOSs
- Ideal Mixtures (Raoult’s Law).

You will need to understand all this material in order to make sense of this course

Chemical thermodynamics concerns the movement of **material** between phases (or chemical species).

Our aim is to establish the driving force for the movement of material and hence a criterion for **equilibrium**.

We will look at:

1. Phase equilibria in more difficult multicomponent systems – **real mixtures**.
2. How computer packages (e.g. Unisim) calculate properties and equilibrium positions in mixtures.

We will cover:

Solid Liquid Equilibrium (SLE)

Vapour Liquid Equilibrium (VLE), including bubbles and drops

Liquid Liquid Equilibrium (LLE), including osmotic equilibrium

Vapour Liquid Liquid Equilibrium (VLLE)

These are essential for separation processes (e.g. distillation, osmosis, liquid-liquid extraction) and reactor design.

Why do you need to know this stuff?

Calculations done on computer - Unisim

Why understand?

1. Critically assess accuracy.
2. Gain an intuitive feel for what is important.

Chemical products often involve chemical equilibrium. Usually these situations are too complex and/or too new to be well modelled by computer. You will have to rely on understanding and intuition.

“An expert is someone who knows some of the worst mistakes that can be made in his subject and how to avoid them.”

Werner Heisenberg, Nobel prize for physics, 1932; head of the German atomic bomb programme.

“An expert is a man who has made all the mistakes which can be made in a very narrow field.”

Niels Bohr, Nobel prize for physics, 1922; a leader of the US atomic bomb programme.



Niels Bohr Archive, BO33; 1936, Copenhagen

A reminder of notation

V Total volume of the system

$\underline{V}$ Molar volume of material in the system. The system may be a pure material or a mixture.
Note: instead of V_m we use $\underline{V}$ in this course

$\underline{V}_i$ Molar volume of pure i

$\overline{V}_i$ Partial molar volume of component i in a mixture

Defined by $$\overline{V}_i = \left(\frac{\partial V}{\partial n_i} \right)_{T,P,n_j(i \neq j)}$$

N.b. $\overline{V}_i = \underline{V}_i$ for a pure substance



e.g. consider a bottle of whiskey

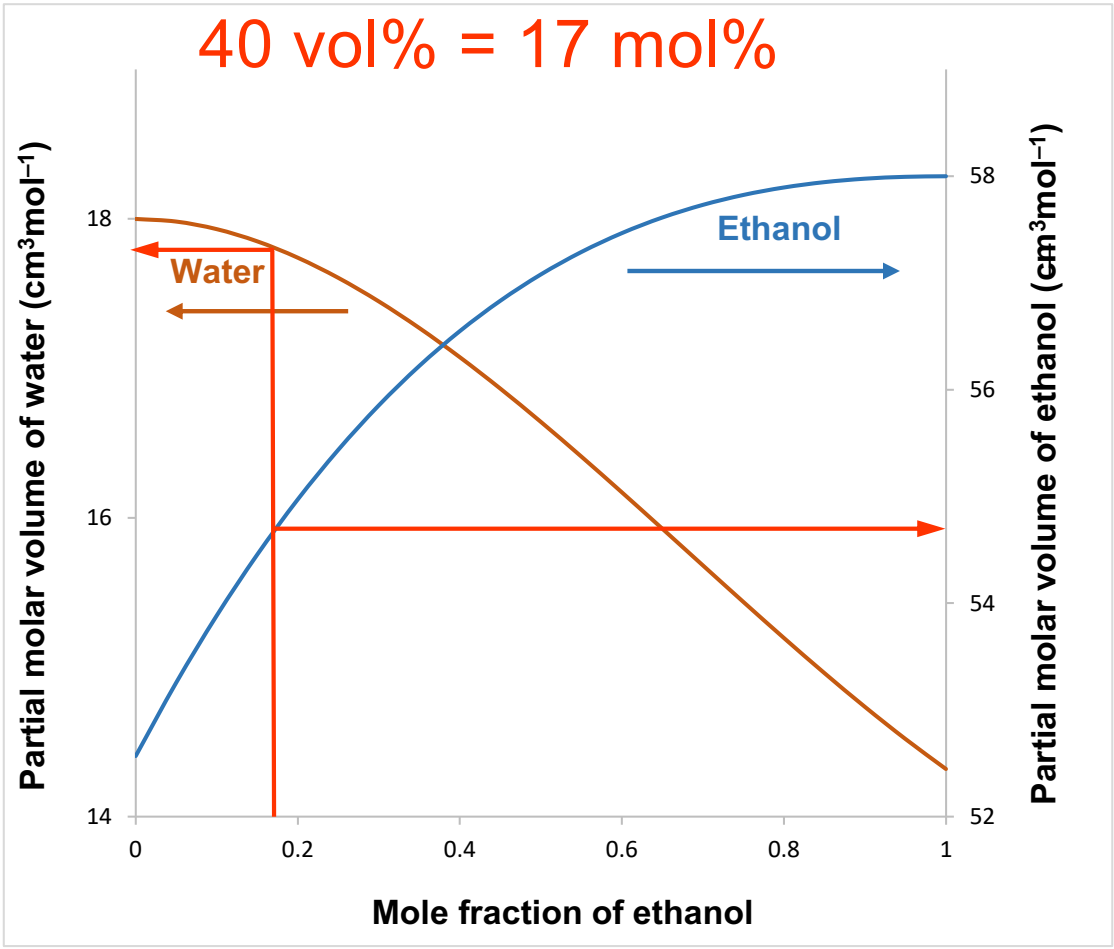
$V = 700\ \mathrm{cm}^3$

$\underline{V}_{H_2O} = 18\ \mathrm{cm}^3\mathrm{mol}^{-1}$

$\underline{V}_{C_2H_5OH} = 58\ \mathrm{cm}^3\mathrm{mol}^{-1}$

$\overline{V}_{C_2H_5OH} = 54.7\ \mathrm{cm}^3\mathrm{mol}^{-1}$

$\overline{V}_{H_2O} = 17.8\ \mathrm{cm}^3\mathrm{mol}^{-1}$



Partial Molar Properties

Particularly important to understand

Chemical potential is partial molar Gibbs energy

If you do not understand the relationship between chemical potential and Gibbs energy:

1. **Look at your Process Calculations notes**
2. **Ask your supervisor**

In the handout, I give a couple of ways of thinking about partial molar properties, which might help you

I won't go through these now

Revision from Process Calculations (page 8)

13 things you need to know:

First Law: $dG = -SdT + VdP$ (1)

for a **closed**, **homogenous** system

Second Law: S = maximised in the universe

$\Rightarrow$ for a closed system at constant T, P

G = minimised (2)

(1) + (2) = all theory, but need EOS to calculate properties

e.g. volume, enthalpy, entropy

An EOS gives $\underline{V}$ as a fn(T,P) {and composition for a mixture}

First EOS $P\underline{V} = RT$ no liquids!

Van der Waals EOS $\left(P + \frac{a}{\underline{V}^2}\right)(\underline{V} - b) = RT$ (3)

1873 – predicts the existence of liquids

All subsequent EOSs based on VdW
(which is a cubic equation)

Use EOS $\Rightarrow$ properties, e.g.

$$\underline{H}(T,P) - \underline{H}^{IG}(T) = \int_{P=0}^{P} \left\{ \underline{V} - T\left(\frac{\partial \underline{V}}{\partial T} \right)_P \right\} dP$$

Pure substance phase equilibrium $\underline{G}^L = \underline{G}^V$ (4)

e.g. VLE

Mixture equilibrium $\left(\frac{\partial G^L}{\partial n_i}\right)_{T,P,n_j} = \left(\frac{\partial G^V}{\partial n_i}\right)_{T,P,n_j}$

$$\mu_i^L = \mu_i^V \quad (5)$$

Chemical potential is the key concept

Chemical potential is a partial molar property:

$$G = \sum \mu_i n_i$$

$$dG = \sum \mu_i dn_i \qquad (6)$$

If we can calculate the chemical potentials of each component in each phase present,
then we can find the equilibrium position

Gibbs-Duhem equation $\sum n_i d\mu_i = 0$ (T, P constant) (7)

Chemical potentials in a mixture are not independent

This is best thought about by **analogy** with volume



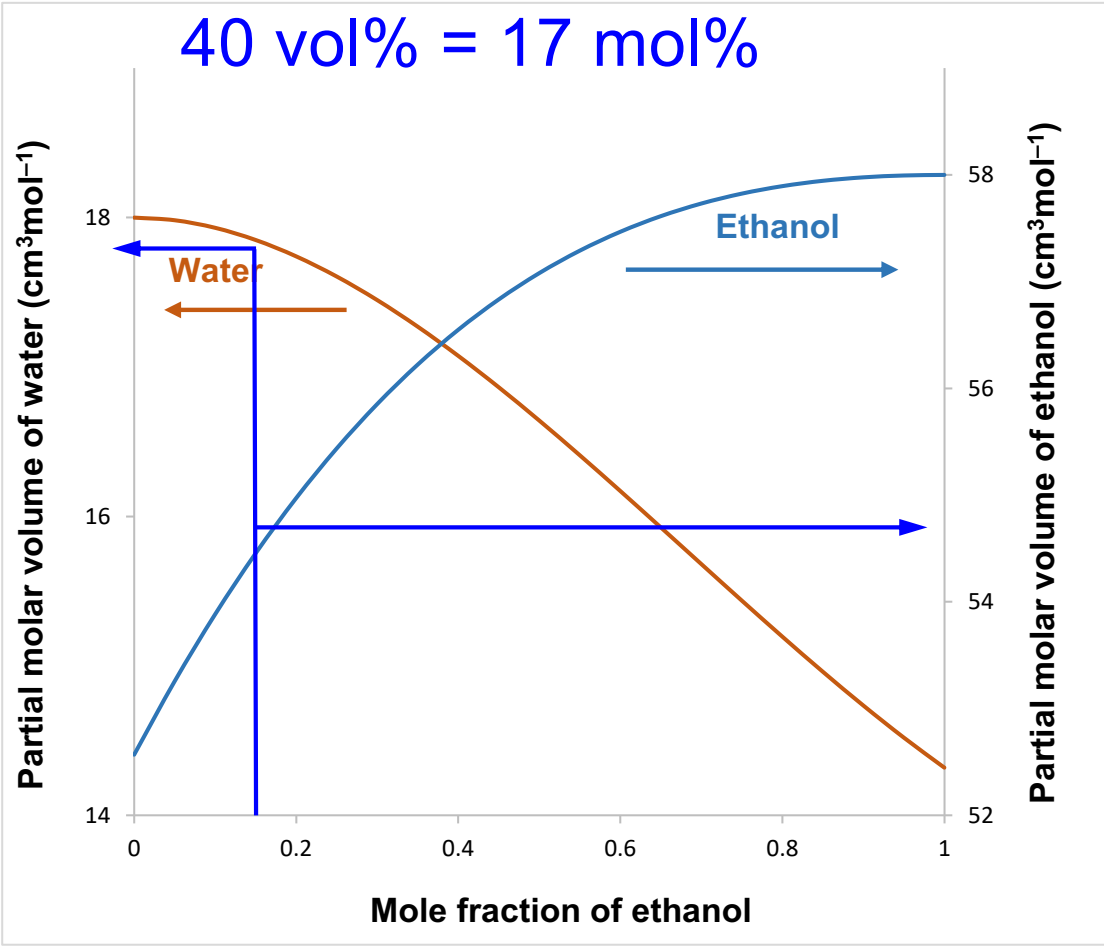
e.g. remember the bottle of whiskey (page 3)

$V = 700\ \mathrm{cm}^3$

$\underline{V}_{H_2O} = 18\ \mathrm{cm}^3\mathrm{mol}^{-1}$

$\underline{V}_{C_2H_5OH} = 58\ \mathrm{cm}^3\mathrm{mol}^{-1}$

40 vol% = 17 mol%



$\overline{V}_{C_2H_5OH} = 54.7\ \mathrm{cm}^3\mathrm{mol}^{-1}$

$\overline{V}_{H_2O} = 17.8\ \mathrm{cm}^3\mathrm{mol}^{-1}$

G-D $\Rightarrow \overline{V}_{H_2O}$ increases when $\overline{V}_{C_2H_5OH}$ decreases and vice versa

How much water and ethanol goes into a 700 cm^3 bottle?

How much water and ethanol goes into a 700 cm^3 bottle?

$$\overline{V}_{C_2H_5OH} = 54.7\ cm^3 mol^{-1}$$

$$\overline{V}_{H_2O} = 17.8\ cm^3 mol^{-1}$$

40 vol% = 17 mol%

$$n_{C_2H_5OH}\overline{V}_{C_2H_5OH} + n_{H_2O}\overline{V}_{H_2O} = nx_{C_2H_5OH}\overline{V}_{C_2H_5OH} + nx_{H_2O}\overline{V}_{H_2O} = V$$

$$n\left(0.17\times 54.7 + 0.83\times 17.8\right) = 700 \Rightarrow n = 29.08 \text{ moles}$$

$$n_{C_2H_5OH} = 0.17\times 29.08 = 4.94 \text{ moles}$$

$$n_{H_2O} = 0.83\times 29.08 = 24.13 \text{ moles}$$

$$\underline{V}_{H_2O} = 18\ cm^3 mol^{-1}$$

$$\underline{V}_{C_2H_5OH} = 58\ cm^3 mol^{-1}$$

$$V_{C_2H_5OH} = 4.94\times 58 = 287\ cm^3$$

$$V_{H_2O} = 24.13\times 18 = 434\ cm^3$$

$287 + 434 = 721\ cm^3$ of alcohol + water make $700\ cm^3$ of whiskey

Calculating μ_i

Pure IG $\quad \mu_i = \underline{G}_i = \underline{G}_i^0 + RT \ln \dfrac{P}{P^0}$

IG $\Rightarrow$ no interactions $\Rightarrow$ same μ_i if in mixture

IG mixture $\quad \mu_i = \underline{G}_i^0 + RT \ln \dfrac{P_i}{P^0}$

$$\mu_i = \underline{G}_i^0 + RT \ln \frac{Px_i}{P^0}$$

$$\mu_i = \underline{G}_i + RT \ln x_i \qquad (8)$$

pure IG at same T, P as mixture

IG mixture $\underline{G} = \sum \mu_i x_i$

$$\underline{G} = \sum \underline{G}_i x_i + RT \sum x_i \ln x_i \qquad (9)$$

pure IG at same T, P as mixture

Gibbs energy of mixing for IG mixture

Not a function of P

Ideal mixture

Define IM such that **mixing properties** are the same as those of a mixture of ideal gases having the same composition

So $$\mu_i = \underline{G}_i + RT \ln x_i \qquad (10)$$

Pure real i at same T, P as mixture

and $$\underline{G} = \sum x_i \mu_i$$

$$\underline{G} = \sum \underline{G}_i x_i + RT \sum x_i \ln x_i \qquad (11)$$

Ideal Gibbs energy of mixing

Not a function of P

Using EOSs to Calculate μ_i

Real pure substance $$\mu_i = \underline{G}_i = \underline{G}_i^0 + \int_{P^0}^{P} \underline{V}_i dP \qquad (12)$$

Real pure i at T, P^0

integrating along an isotherm at T

EOS $\Rightarrow$ $\underline{V}_i$ as fn (T, P)

Can convert real $\leftrightarrow$ ideal by integrating to $P = 0$

Real mixture (at T, P)

$$\underline{G} = \sum \mu_i x_i = \underline{G}^0 + \int_{P^0}^{P} \underline{V} dP$$

mixture at T, P^0

and

$$\underline{G}^0 = \sum \mu_i^0 x_i$$

component i in mixture of same composition at T, P^0

and

$$\underline{V} = \sum \overline{V}_i x_i$$

molar volume of mixture

partial molar volume of component i in mixture

so

$$\mu_i = \mu_i^0 + \int_{P^0}^{P} \overline{V}_i dP \qquad (13)$$

Care: non-zero ideal Gibbs energy of mixing even at $P = 0$

Mixture EOS $\Rightarrow$ $\overline{V}_i$ as fn(composition, T, P)

Only possible for "easy" cases: homologues
gases
isomers